

§10.

1000.10.1

$y = \int_0^x e^{-t} dt$, y 轴 $x \rightarrow +\infty$ 渐近线 面积.

[solution]

$$1) \lim_{x \rightarrow +\infty} \int_0^x e^{-t} dt = \lim_{x \rightarrow +\infty} \int_0^x e^{-u} du = \lim_{x \rightarrow +\infty} (e^{-u} + e^{-u}) \Big|_0^x$$

$$2) S = \lim_{x \rightarrow +\infty} \left(2 \int_0^x (xe^{-t} + e^{-t}) dx \right) = \lim_{x \rightarrow +\infty} 2 - 2 \int_0^x (xe^{-t} + e^{-t}) dx = 2$$

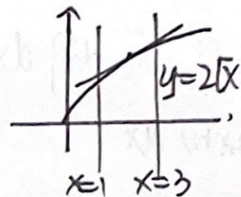
$$= \lim_{x \rightarrow +\infty} 4 \left[(u^2 + u)(-e^{-u}) \Big|_0^x + \int_0^x e^{-u} d(u^2 + u) \right]$$

$$= 4 \left[(u^2 + u)(-e^{-u}) \Big|_0^{+\infty} + \int_0^{+\infty} (2u + 1) d(-e^{-u}) \right]$$

$$\therefore = 12$$

1000.10.2

$y = 2\sqrt{x}$ 与其上点 $(t, 2\sqrt{t})$ 处切线 以及 $x=1$, $x=3$ 围成面积 记为 $A(t)$. ($t > 0$). $A(t)$ 何时 切线?



$$y = 2\sqrt{x} \quad y' = x^{-\frac{1}{2}}$$

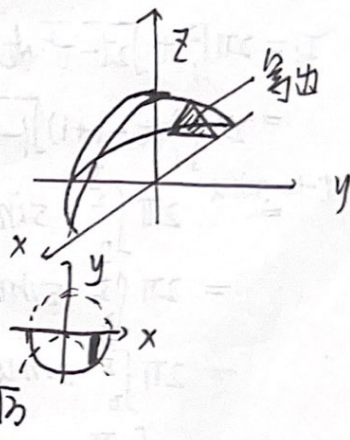
$$\text{切线 } y - 2\sqrt{t} = t^{-\frac{1}{2}}(x - t)$$

$$\Rightarrow y = \frac{x}{\sqrt{t}} + \sqrt{t} \quad t \in [1, 3].$$

$$S = \frac{1}{2} \left(\frac{2}{\sqrt{t}} + \sqrt{t} + \frac{1}{\sqrt{t}} + \sqrt{t} \right) = 2\sqrt{t} + \frac{4}{\sqrt{t}} = 2t^{\frac{1}{2}} + 4t^{-\frac{1}{2}}$$

$$S'(t) = t^{-\frac{1}{2}} - 2t^{-\frac{3}{2}} \stackrel{!}{=} 0 \Rightarrow t = 2.$$

$$\text{切线: } y = \frac{\sqrt{2}}{2}x + \sqrt{2}$$



1000.10.3

$$y = x^2 - 1 \text{ 与 } y = 0$$

$$V = 2 \int_0^1 \frac{\sqrt{3}}{4} (1 - x^2)^2 dx$$

$$= \frac{\sqrt{3}}{2} \int_0^1 (1 + x^4 - 2x^2) dx = \frac{4}{15} \sqrt{3}$$

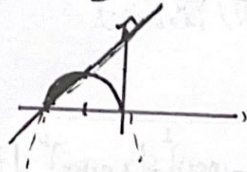
1000.10.4

$$y = \sqrt{2x - x^2} \text{ 与 } y = x. \text{ 所围 绕 } x=2. \text{ 体积}$$

[solution]

$$\text{令 } t = x - 2 \quad x = t + 2$$

$$\rightarrow y = \sqrt{-t(t+2)} \text{ 与 } y = t+2 \quad t \in (-2, 0)$$



$$V = \int_{-2}^0 2\pi \left[(t+2) - \sqrt{-t(t+2)} \right] dt$$

$$= 2\pi \int_{-2}^0 t \left[(2-t) - \sqrt{t(2-t)} \right] dt$$

$$= 2\pi \int_1^2 t \left[\sqrt{t(2-t)} - (2-t) \right] dt$$

$$= 2\pi \int_1^2 t \sqrt{2t-t^2} dt - 2\pi \int_1^2 (t-1) dt$$

$$= I_1 - I_2 \quad I_2 = \frac{4}{3}\pi$$

$$I_1 = 2\pi \int_1^2 t \sqrt{2t-t^2} dt$$

$$= 2\pi \int_1^2 (t-1+1) \sqrt{1-(t-1)^2} d(t-1)$$

$$\text{令 } t-1 = \sin u \quad 2\pi \int_0^{\frac{\pi}{2}} (\sin u + 1) \sqrt{1-\sin^2 u} \cdot \cos u du$$

$$= 2\pi \int_0^{\frac{\pi}{2}} (\sin u + 1) \cos^2 u du \quad (1-\sin^2 u)$$

$$= 2\pi \int_0^{\frac{\pi}{2}} \sin u + 1 - \sin^2 u - \sin^3 u du$$

$$= 2\pi \left[\frac{\pi}{2} + 1 - \frac{1}{2} \frac{\pi}{2} - \frac{2}{3} \cdot 1 \right]$$

$$= \frac{\pi^2}{2} + \frac{2}{3}\pi$$

$$\Rightarrow I = \frac{\pi^2}{2} + \frac{2}{3}\pi - \frac{4}{3}\pi = \frac{\pi^2}{2} - \frac{2}{3}\pi$$

1000.10.5

$$\text{曲线 } \begin{cases} x = 3(t - \sin t) \\ y = 3(1 - \cos t) \end{cases} \quad (0 \leq t \leq 2\pi) \text{ 所围面积.}$$

$$I = \int_0^{2\pi} \sqrt{(x'_t)^2 + (y'_t)^2} dt = \int_0^{2\pi} [3(1 - \cos t)]^2 + [3 \sin t]^2 dt$$

$$= 3 \int_0^{2\pi} \sqrt{1 + \cos^2 t + \sin^2 t - 2\cos t} dt = 3\sqrt{2} \int_0^{2\pi} \sqrt{1 - \cos t} dt$$

$$= 6 \int_0^{2\pi} \sqrt{1 - \sin^2 \frac{t}{2}} dt = 12 \int_0^{\pi} \sqrt{1 - \sin^2 u} du$$

$$= 12 \int_0^{\pi} \cos u du = 0.24$$

1000.10.6

$$\text{曲线 } r = 1 + \cos \theta, \quad 0 \leq \theta \leq \pi \text{ 所围为}$$

[solution].

$$I = \int_0^{\pi} \sqrt{(1+\cos \theta)^2 + \sin^2 \theta} d\theta = \int_0^{\pi} \sqrt{2+2\cos \theta} d\theta$$

$$= \sqrt{2} \int_0^{\pi} \sqrt{1 + \cos^2 \frac{\theta}{2} - \sin^2 \theta} d\theta = \sqrt{2} \int_0^{\pi} \cos \frac{\theta}{2} d\theta$$

$$= 2 \sin \frac{\theta}{2} \Big|_0^{\pi} = 4$$

1000.10.7

$$f(x) = \int_x^1 \cos t^2 dt \text{ 在区间 } [0,1] \text{ 上平均值}$$

$$I = \frac{\int_0^1 f(x) dx}{1-0} = \int_0^1 f(x) dx$$

$$f(x) = \int_x^1 \cos t^2 dt \neq I = \int_0^1 \left(\int_x^1 \cos t^2 dt \right) dx$$

$$I = \int_0^1 \left(\int_x^1 \cos(x+1-t)^2 dt \right) dx$$

$$= \int_0^1 \left[\cos(x+1) \int_x^1 \cos t dt - \sin(x+1) \int_x^1 \sin t dt \right] dx$$

$$= \frac{1}{2} \int_0^1 \sin(x+2) dx - \int_0^1 \sin(2x+1) dx$$

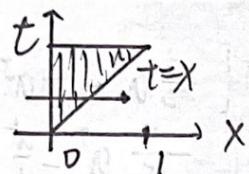
$$= \frac{1}{2} \sin 1 + \frac{1}{2} \sin 3 - \sin 2$$

$$I = \int_0^1 \int_x^1 \cos t^2 dt dx$$

$$= \int_0^1 \int_0^{x^2} \cos t^2 dx dt$$

$$= \int_0^1 t \cos t^2 dt = \frac{1}{2} \int_0^1 \cos t^2 dt^2$$

$$= \frac{1}{2} \int_0^1 \cos u du = \frac{1}{2} \sin u \Big|_0^1 = \frac{1}{2} \sin 1$$



1000.10.8

$v = 3t^2 + 2t$ m/s. 直线运动

在 $t=0$ s 到 $t=8$ s. 这段时间内的平均速度

(solution)

$$\left(\int_0^8 3t^2 + 2t dt \right) / 8 = \frac{t^3 + t^2}{3} \Big|_0^8 = \frac{27+9}{3} = 12$$

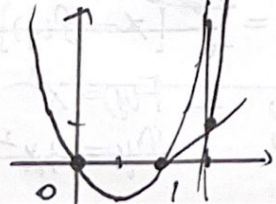
1000.10.9

过 $(\frac{3}{2}, \frac{1}{2})$ 作 $y = x^2 - x$ 的切线. 求该曲线与切线围成图形的面积

$$y' = 2x_0 - 1 \quad \frac{1}{2} - y_0 = (2x_0 - 1)(\frac{3}{2} - x_0)$$

$$\frac{3}{2} - (x_0^2 - x_0) = (2x_0 - 1)(\frac{1}{2} - x_0)$$

$$\Rightarrow (x, y) = (2, 2) \text{ 或 } (1, 0)$$



$$\begin{cases} y = x^2 - x \\ y = x + 1 \end{cases}$$

$$S = \int_1^{\frac{3}{2}} (x^2 - x) + (-x + 1) dx + \int_{\frac{3}{2}}^2 (x^2 - x) + (-3x + 4) dx$$

$$= \left[\frac{x^3}{3} - x^2 + x \right]_1^{\frac{3}{2}} + \left[\frac{x^3}{3} - x^2 - 3x + 4 \right]_{\frac{3}{2}}^2 = \frac{1}{24} + \frac{1}{24} = \frac{1}{12}$$

1000.10.10

$y = ax^2 + bx + c$ 通过 $(0,0)$ 与 $(1,2)$ 两点, 其中 $a < -2$. 求 a, b, c s.t. 与 $y = -x^2 + 2x = x(2-x)$ 所围 S. min.

(solution)

$$0 = 0 + 0 + c \Rightarrow c = 0$$

$$2 = a + b + 0 \Rightarrow a + b = 2$$

$$\frac{-b}{2a} > 4 \quad b > 4 \quad a < -2 \quad \frac{-b}{2a} < -4 = \frac{a-2}{2a} = \frac{1}{2} - \frac{1}{a} \in (\frac{1}{2}, 1)$$

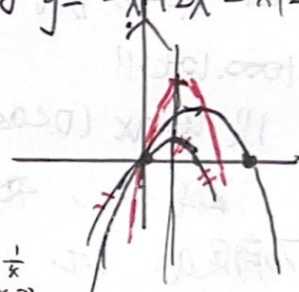
$$\Rightarrow \begin{cases} y = ax^2 + (2-a)x \\ y = -x^2 + 2x \end{cases} \quad \begin{cases} (a+1)x^2 - ax = 0 \\ x((a+1)x - a) = 0 \end{cases}$$

$$x_1 = 0 \quad x_2 = \frac{a}{a+1}$$

$$S = \int_0^{\frac{a}{a+1}} [ax^2 + (2-a)x] - [-x^2 + 2x] dx$$

$$= \left[\frac{a}{a+1} \left[(a+1)x^2 - ax \right] \right]_0^{\frac{a}{a+1}} = \frac{a+1}{3} x^3 - \frac{a}{2} x^2 \Big|_0^{\frac{a}{a+1}}$$

$$= -\frac{1}{6} \frac{a^3}{(a+1)^2}$$



$$S = -\frac{1}{6} \frac{a^3}{(a+1)^2}$$

$$S'_a = -\frac{1}{6} \frac{3a^2(a+1)^2 - a^3 \cdot 2(a+1)}{(a+1)^4}$$

$$= -\frac{1}{6} \frac{3a^2(a+1) - 2a^3}{(a+1)^3} = -\frac{1}{6} \frac{3a^2 + 3a^2 - 2a^3}{(a+1)^3}$$

$$= -\frac{1}{6} \frac{a^2 + 3a^2}{(a+1)^3} = -\frac{1}{6} \frac{a^2(a+3)}{(a+1)^3} \quad a \in (-\infty, -2)$$

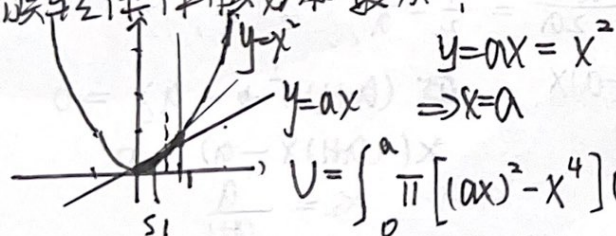
$$\sum S'_a = 0 \Rightarrow a = -3 \rightarrow b = 5.$$

$$\Rightarrow a = -3, b = 5, c = 0.$$

1000. 10t. 11

设 $y = ax$ ($0 < a < 1$) 与抛物线 $y = x^2$

围成 S_1 , 相交后部分与 $x=1$ 围成 S_2 .
求定 a , s.t. S_1 与 S_2 绕 x 轴一周所得旋
族体积之和最小?



$$V = \int_0^a \pi [(ax)^2 - x^4] dx + \int_a^1 \pi [x^4 - (ax)^2] dx$$

$$V = \pi \left[\frac{a^2}{3} x^3 - \frac{x^5}{5} \Big|_0^a + \frac{x^5}{5} - \frac{a^2}{3} x^3 \Big|_a^1 \right]$$

$$= \pi \left[\left(\frac{a^5}{3} - \frac{a^5}{5} \right) + \left(\frac{1}{5} - \frac{a^2}{3} \right) - \left(\frac{a^5}{5} - \frac{a^5}{3} \right) \right]$$

$$= \pi \left[\frac{2}{3} a^5 - \frac{2}{5} a^5 + \frac{1}{5} - \frac{a^2}{3} \right]$$

$$= \pi \left[\frac{4}{15} a^5 - \frac{a^2}{3} + \frac{1}{5} \right]$$

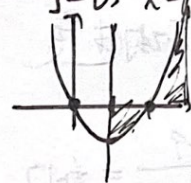
$$S' = \pi \left[\frac{4}{3} a^4 - \frac{2}{3} a \right] = \frac{2}{3} \pi a [2a^3 - 1] \quad a \in (0, 1).$$

$$2a^3 = 1 \quad a = \left(\frac{1}{2}\right)^{\frac{1}{3}}$$

1000. 10t. 12

求曲线 $y = x^2 - 2x$ ($1 \leq x \leq 3$).

$y=0, x=1$ 与 $x=3$. $\rightarrow S$ 绕 y 轴, 体积

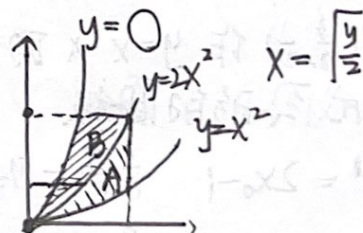


$$V = \int_1^2 2\pi x (0 - (x^2 - 2x)) dx$$

$$+ \int_2^3 2\pi x (x^2 - 2x - 0) dx$$

1000. 10t. 13

10. L. 求 $y = 2x^2$



$$S = \int_0^x (2x^2 - x^2) dx = \int_0^x x^2 dx = \frac{x^3}{3}$$

$$S = \frac{x^3}{3} = \int_0^{2x^2} \left[\frac{y}{2} - \phi(y) \right] dy = \int_0^{2x^2} \left[\frac{y}{2} - \phi(y) \right] dy$$

$$\frac{x^3}{3} = \frac{1}{3} y^{\frac{3}{2}} - \frac{F(y)}{2x^2} = \frac{4}{3} x^3 - \frac{F(y)}{2x^2} \rightarrow F(y) = x^3$$

$$\phi(y) = 2x^2$$

$$\int_0^{2x^2} \left[\frac{y}{2} - f(y) \right] dy = \frac{x^3}{3} \Rightarrow \frac{4}{3}x^3 - F(2x^2) = \frac{x^3}{3}$$

$$\begin{cases} F(2x^2) = -\frac{x^3}{3} & f(2x^2) = -x^2 \cdot (4x) = -4x^3 = -\sqrt{2}(2x^2)^{3/2} \\ f(x) = -\sqrt{2}(2x)^{3/2} \end{cases}$$

$$\left[\frac{2x^2}{2} - f(2x^2) \right] (4x) = x^2 \Rightarrow f(2x^2) = \frac{3}{4}x \quad \begin{cases} 2x^2 = t \\ t = \frac{x}{2} \end{cases} \quad f(t) = \frac{3\sqrt{t}}{8}, \quad x = \frac{3\sqrt{2y}}{8}$$

$$\begin{aligned} 12. \quad V &= \int_0^x 2\pi x \cdot (2x^2 - x^2) dx \\ &= \int_0^x 2\pi x^3 dx = \frac{\pi x^4}{2} \end{aligned}$$

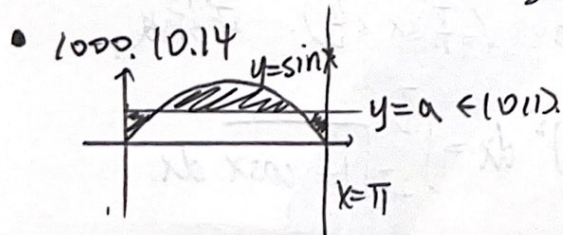
$$V = \frac{\pi x^4}{2} = \int_0^{2x^2} \pi \left[\frac{y}{2} - f^2(y) \right] dy$$

$$\frac{\pi x^4}{2} = \int_0^{2x^2} \pi \left[\frac{y}{2} - f^2(y) \right] dy$$

$$\xrightarrow{x \neq 0} 2\pi x^3 = 4\pi \left(\frac{1}{2}x^2 - f^2(2x^2) \right)$$

$$\Rightarrow f^2(2x^2) = \frac{x^2}{2} \quad f(2x^2) = \frac{x}{\sqrt{2}}$$

$$t = 2x^2 \rightarrow f(t) = \frac{\sqrt{t}}{2} \quad x = \frac{\sqrt{y}}{2}$$



$$\begin{aligned} V &= \int_0^\pi \pi |a^2 - \sin^2 x| dx \quad \begin{cases} a = \sin x \\ x = \arcsin a \end{cases} \\ &= 2\pi \left[\int_0^{\arcsin a} (a^2 - \sin^2 x) dx + \int_{\arcsin a}^\pi (\sin^2 x - a^2) dx \right] \\ &= 2\pi \left[\frac{a^2 x}{2} - \frac{x}{2} + \frac{2 \sin x \sqrt{1 - \sin^2 x}}{4} \Big|_0^{\arcsin a} + \left(\frac{x}{2} - a^2 x + \frac{2 \sin x \sqrt{1 - \sin^2 x}}{4} \right) \Big|_{\arcsin a}^\pi \right] \\ &= 2\pi \left[(a^2 - \frac{1}{2}) \arcsin a + \frac{a \sqrt{1 - a^2}}{2} + (\frac{1}{2} - a^2) (\pi - \arcsin a) + \frac{a \sqrt{1 - a^2}}{2} \right] \\ &= 2\pi \left[a \sqrt{1 - a^2} - (a^2 - \frac{1}{2}) \frac{\pi}{2} \right] \\ &= 2\pi a \sqrt{1 - a^2} - \pi^2 (a^2 - \frac{1}{2}) \end{aligned}$$

$$V'_a = 2\pi \sqrt{1 - a^2} - 2\pi a^2 (1 - a^2)^{-\frac{1}{2}} - 2\pi a^2 \stackrel{?}{=} 0$$

$$(4 + \pi^2) a^4 - (4 + \pi^2) a^2 + 1 = 0$$

$$a^2 = t \quad (4 + \pi^2) t^2 - (4 + \pi^2) t + 1 = 0$$

$$t = \frac{1}{2} - \frac{1}{2} \sqrt{\frac{\pi}{4 + \pi^2}}$$

$$= (2a^2 - 1) \arcsin a + a \sqrt{1 - a^2} + \frac{\pi}{2} \left(\frac{1}{2} - a^2 \right)$$

$$\begin{aligned} V'_{|a|} &= 4a \cdot \arcsin a - (2a^2 - 1) \frac{1}{\sqrt{1 - a^2}} + \sqrt{1 - a^2} \\ &\quad + \frac{a^2}{\sqrt{1 - a^2}} - \pi a \stackrel{?}{=} 0 \end{aligned}$$

$$V'_{|a|} = 4a \cdot \arcsin a - (a^2 - 1) \frac{1}{\sqrt{1 - a^2}} + \sqrt{1 - a^2} - \pi a = 0$$

$$V'(a) = 4a \operatorname{arcsinh} a - \pi a + 2\sqrt{1+a^2} \cdot 0 = 0$$

$$\begin{aligned} V &= 2 \left[\int_0^{\operatorname{arcsinh} a} \pi (a - \sin x)^2 dx + \int_{\operatorname{arcsinh} a}^{\frac{\pi}{2}} \pi (\sin x - a)^2 dx \right] \\ &= 2\pi \left[\int_0^{\operatorname{arcsinh} a} (a - \sin x)^2 dx + \int_{\operatorname{arcsinh} a}^{\frac{\pi}{2}} (\sin x - a)^2 dx \right] \\ &= 2\pi \left[\left(\frac{1}{2} + a^2 \right) \frac{\pi}{2} + 2a \cos x \Big|_0^{\frac{\pi}{2}} - \frac{\sin 2x}{4} \Big|_0^{\frac{\pi}{2}} \right] \\ &= 2\pi \left[\left(\frac{\pi}{2} \right) \left(\frac{1}{2} + a^2 \right) - 2a \right] \\ &= \pi a^2 + \frac{\pi^2}{2} - 4\pi a \end{aligned}$$

$$V'(a) = 2\pi^2 a - 4\pi \rightarrow a = \frac{2}{\pi}. \quad V'(a) > 0$$

$$\therefore V\left(\frac{2}{\pi}\right) = \frac{1}{2}\pi - 4 \quad \text{极小值}$$

$$\begin{aligned} V(0) &= \frac{1}{2}\pi^2 \\ V(1) &= \frac{3}{2}\pi^2 - 4\pi \end{aligned} \rightarrow V(0) = V_{\max} = \frac{1}{2}\pi^2$$

1000. 10.15

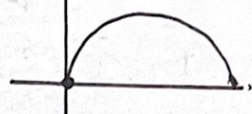
$$y = \frac{1}{x^2+1} \quad \text{与 } x \text{ 轴之间 } S.$$

$$\begin{aligned} S &= 2 \int_0^{+\infty} \frac{1}{x^2+1} dx \\ &= 2 \arctan x \Big|_0^{+\infty} \\ &= \pi \end{aligned}$$

1000. 10.16.

$$\begin{cases} x = 2(t - \sin t) \\ y = 2(1 - \cos t) \end{cases}$$

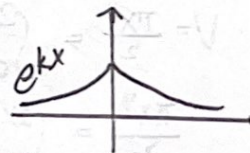
求 $0 \leq t \leq 2\pi$ 的一弧与 x 轴.
位于 I 段上.



$$\begin{aligned} S &= \int_0^{2\pi} 2(1 - \cos t) \cdot 2(1 - \cos t) dt \\ &= \int_0^{2\pi} 4(1 - \cos t)^2 dt \\ &= \int_0^{2\pi} 4(1 - 2\cos t + \cos^2 t) dt \\ &= \int_0^{2\pi} 4 \left(1 - 2\cos t + \frac{1 + \cos 2t}{2} \right) dt \\ &= \int_0^{2\pi} (3 - 2\cos t + 2\cos^2 t) dt \\ &= 3t - 2\sin t + 2 \int_0^{2\pi} \cos^2 t dt \\ &= 3t - 2\sin t + 2 \int_0^{2\pi} \frac{1 + \cos 2t}{2} dt \\ &= 3t - 2\sin t + \int_0^{2\pi} (1 + \cos 2t) dt \\ &= 3t - 2\sin t + \left(t + \frac{\sin 2t}{2} \right) \Big|_0^{2\pi} \\ &= 12\pi \end{aligned}$$

1000. 10.17

$$k > 0, \quad f(x) = \begin{cases} e^{kx}, & x \leq 0 \\ e^{-kx}, & x > 0. \end{cases}$$



$$S = 2 \int_0^{+\infty} e^{-kx} dx = \left[-\frac{2}{k} e^{-kx} \right]_0^{+\infty} = \frac{2}{k}$$

$$S'_k = -\frac{2}{k^2} < 0. \quad \therefore S \downarrow$$

1000. 10.18

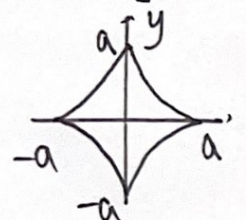
$$y = \int_{-\frac{\pi}{2}}^x \sqrt{\cos t} dt \quad \left(-\frac{\pi}{2} \leq x \leq \frac{\pi}{2} \right) \quad \text{求弧长}$$

$$l = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{1 + (y'(x))^2} dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{1 + \cos x} dx$$

$$\begin{aligned}
 I &= 2 \int_0^{\frac{\pi}{2}} \sqrt{1 + \cos 2 \cdot \frac{x}{2}} dx = 2 \int_0^{\frac{\pi}{2}} \sqrt{2 - 2 \sin \frac{x}{2}} dx \\
 &= 2 \int_0^{\frac{\pi}{2}} \sqrt{2} \cos \frac{x}{2} dx = 4 \sqrt{2} \sin \frac{x}{2} \Big|_0^{\frac{\pi}{2}} = 4 \sqrt{2} \cdot \frac{\sqrt{2}}{2} = 4
 \end{aligned}$$

1000. 10.19.

$x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ 绕 x -轴一周, 旋转体体积.

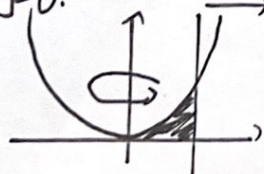
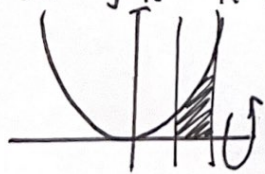


$$\begin{aligned}
 V &= 2 \int_0^a \pi y^2 dx \\
 &= 2 \int_0^a \pi (a^{\frac{2}{3}} - x^{\frac{2}{3}})^3 dx \\
 &= 2\pi \int_0^a a^2 + (1 - 3a^{\frac{4}{3}})x^{\frac{2}{3}} + 3a^{\frac{2}{3}}x^{\frac{4}{3}} dx \\
 &= 2\pi \left[a^2 x + \frac{3}{5}(1 - 3a^{\frac{4}{3}})x^{\frac{5}{3}} + \frac{3}{7}a^{\frac{2}{3}}x^{\frac{7}{3}} \right]_0^a \\
 &= 2\pi \left[a^3 + \frac{3}{5}(1 - 3a^{\frac{4}{3}})a^{\frac{5}{3}} + \frac{9}{7}a^{\frac{2}{3}}a^{\frac{7}{3}} \right] \\
 &= \dots \frac{27}{10} \pi a^3
 \end{aligned}$$

1000. 10.20

D1: $y=x^2$, $x=1$ $x=2$ $y=0$. $\frac{x}{y}$ V_1

D2: $y=x^2$ $x=1$ $y=0$. $\frac{y}{x}$ V_2



$$V_1 = \int_1^2 \pi (x^2)^2 dx = \frac{2}{3} \pi \frac{2^3}{3} \pi$$

$$V_2 = \int_0^1 2\pi x \cdot x^2 dx = \frac{\pi}{2}.$$

1000. 10.21

$y=f(x)$ 与 $y=\int_0^{\arcsin x} e^{-t^2} dt$ 在 $(0,1)$ 切线相同.

$$y' = e^{-(\arcsin x)^2} \cdot \left(-\frac{1}{\sqrt{1-x^2}} \right)$$

$$y'(0) = 1 \cdot \left(+\frac{1}{\sqrt{1-0}} \right) = +1$$

$$\begin{aligned}
 (\arcsin x)' &= \frac{1}{\sqrt{1-x^2}} \\
 (\arccos x)' &= -\frac{1}{\sqrt{1-x^2}}
 \end{aligned}$$

$$\text{切: } y-0 = +1(x-0) \rightarrow y=+x$$

$$\lim_{n \rightarrow \infty} n f\left(\frac{1}{n}\right) = \lim_{n \rightarrow \infty} \frac{f\left(\frac{1}{n}\right)}{\frac{1}{n}}$$

$$\stackrel{\frac{1}{n}=t}{=} \lim_{t \rightarrow 0^+} \frac{f(t)}{t} = \lim_{t \rightarrow 0^+} \frac{f(t)-f(0)}{t-0} = f'(0) = 1$$

1000. 10.21

$$\begin{aligned}
 y &= \frac{1}{n} \sin nx \quad \int_0^{2\pi} \left| \frac{\sin nx}{n} \right| dx = \int_0^{2\pi} \left| \frac{\sin t}{n} \right| \frac{1}{n} dt \\
 &= \frac{1}{n^2} \int_0^{2\pi} |\sin t| dt = \frac{2\pi}{n^2} \int_0^{\pi} \sin t dt = \frac{8}{n} \uparrow \downarrow
 \end{aligned}$$

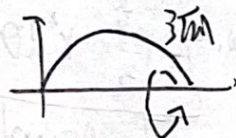
$$S = \int_0^{2\pi} \sqrt{1 + (y')^2} dx = \int_0^{2\pi} \sqrt{1 + (\cos nx)^2} dx$$

$$S = \int_0^{2\pi} \sqrt{1 + \frac{1 + \cos 2t}{2}} dt$$

$$\stackrel{t=n\pi}{=} \int_0^{2n\pi} \sqrt{1 + \cos t} dt = \int_0^{2\pi} \sqrt{1 + \cos t} dt \quad \text{与n无关}$$

1000. 10t. 2

$$\begin{cases} x = a(t - \sin t) \\ y = a(1 - \cos t) \end{cases} \quad t \in [0, 2\pi]$$



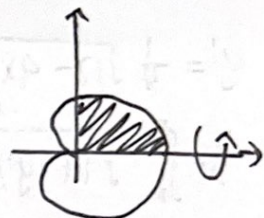
$$\begin{aligned} \text{IV } l &= \int_0^{2\pi} \sqrt{a^2(1 - \cos t)^2 + a^2 \sin^2 t} dt \\ &= a \int_0^{2\pi} \sqrt{1 + \cos^2 t - 2\cos t + \sin^2 t} dt = a \int_0^{2\pi} \sqrt{2 - 2\cos t} dt \\ &= 2a \int_0^{2\pi} \sqrt{1 - \cos \frac{t}{2}} dt = 2a \int_0^{2\pi} \sin \frac{t}{2} dt = +8a \end{aligned}$$

$$\begin{aligned} \text{2. 10t. 1: } S &= 2 \int_0^{2\pi} a(1 - \cos t) \sqrt{a^2(1 - \cos t)^2 + a^2 \sin^2 t} dt \\ &= 2a^2 \int_0^{2\pi} (1 - \cos t)^{3/2} dt = 4a^2 \int_0^{2\pi} \sin^3 \frac{t}{2} dt \\ &= 8a^2 \int_0^{\pi} \sin^3 u du = 16a^2 \cdot \frac{2}{3} = \frac{32}{3}a^2 \end{aligned}$$

$$\rightarrow \frac{16}{3}\pi a^2 = 8a \rightarrow a = \frac{3}{8\pi}$$

1000. 10t. 3

$$r = 2(1 + \cos \theta) \quad \text{和} \quad \theta = 0, \theta = \frac{\pi}{2}$$



$$\begin{aligned} U &= \int_0^{\frac{\pi}{2}} \pi y^2 dx \quad \begin{matrix} y = r \sin \theta \\ x = r \cos \theta = 2(1 + \cos \theta) \cos \theta \end{matrix} \\ &= \int_0^{\frac{\pi}{2}} \pi \cdot 4(1 + \cos \theta)^2 \cdot \sin^3 \theta \cdot [-2 \sin \theta (1 + \cos \theta) + 2 \cos \theta \sin \theta] d\theta \\ &= \int_0^{\frac{\pi}{2}} \pi 4(1 + \cos \theta)^2 \cdot \sin^3 \theta \cdot [2 - 2\cos \theta + 2\cos \theta] d\theta \\ &= 8\pi \int_0^{\frac{\pi}{2}} (1 + \cos \theta)^2 \sin^3 \theta (1 + \cos 2\theta) d\theta \\ &= 8\pi \int_0^{\frac{\pi}{2}} -2\cos^2 \theta (1 + \cos^2 \theta + 2\cos \theta) (1 - \cos^2 \theta) d\cos \theta \\ &\stackrel{\cos \theta = t}{=} 8\pi \int_0^1 +2t^2(1+t^2+2t)(1-t^2) dt \\ &= 8\pi \int_0^1 2t^2 - 2t^6 + 4t^3 - 4t^7 dx = 8\pi \cdot \left[\frac{2}{3} - \frac{2}{7} + 1 - \frac{4}{5} \right] \\ &= 20\pi \end{aligned}$$

1000. 10t. 4. 当 $x > 0$ 时 $y = \frac{1}{4} \int_0^2 x \sqrt{12 - x^2 t^2} dt$ 令

$$12 - x^2 t^2 > 0 \quad x^2 t^2 \leq 12 \quad xt \leq 2\sqrt{3}$$

$$x \leq \frac{2\sqrt{3}}{t} \quad \text{保证 } xt < 12 \text{ 时, } x \leq \sqrt{3}$$

$$\text{令 } xt = u. \quad y = \frac{1}{4} \int_0^{2x} \sqrt{12 - u^2} \frac{du}{x}$$

$$y' = \frac{1}{4} \sqrt{12 - 4x^2} \cdot 2 = \sqrt{3 - x^2}$$

$$S = \int_0^{\sqrt{3}} \sqrt{1 + (y')^2} dx = \int_0^{\sqrt{3}} \sqrt{4 - x^2} dx \quad \frac{\pi}{2}$$

$$\begin{aligned} x &= 2\sin t \quad 2 \int_0^{\frac{\pi}{2}} \cos t \cdot 2 \cos t dt = \frac{2}{4} \int_0^{\frac{\pi}{2}} \frac{1 + \cos 2t}{2} dt \\ &= \frac{\pi}{2} + \frac{\sqrt{3}}{2} \end{aligned}$$

1000.10t.6

$L \quad [-\frac{\pi}{2}, \frac{\pi}{2}]$ 曲线 $L: x \in [-\frac{\pi}{2}, \frac{\pi}{2}] \quad f(x) = 1 + \sin x$

求?

[solution].

$$\bar{x} = \frac{\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} f(x) x dx}{\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} f(x) dx} = \frac{\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x(1 + \sin x) dx}{\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1 + \sin x) dx}$$

1000.10t.5

曲线 $r = 1 + \cos \theta$ 在基点 $(\frac{\pi}{4}, 1 + \frac{\sqrt{2}}{2})$ 处切线及
x 轴所围面积?

[solution].

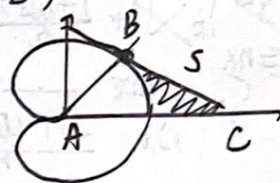
$$\begin{cases} x = r \cos \theta = \cos \theta (1 + \cos \theta) = \cos \theta + \frac{1}{2} + \frac{1}{2} \cos 2\theta \\ y = r \sin \theta = \sin \theta (1 + \cos \theta) = \sin \theta + \frac{1}{2} \sin 2\theta \end{cases}$$

$$y' = \frac{y'_{\theta}}{x'_{\theta}} = \frac{-\sin \theta - \sin 2\theta}{\cos \theta + \cos 2\theta} \bigg|_{\theta = \frac{\pi}{4}} = -\frac{1 - \sqrt{2}}{1 + \sqrt{2}}$$

$$y - (\frac{\sqrt{2}}{2} + \frac{1}{2}) = (1 - \sqrt{2})(x - (\frac{\sqrt{2}}{2} + \frac{1}{2}))$$

$$\rightarrow y = (1 - \sqrt{2})x + 1 + \frac{\sqrt{2}}{2}$$

$$S_{ABC} = \frac{1}{2} x_B \cdot y_B = \frac{10 + 7\sqrt{2}}{8}$$



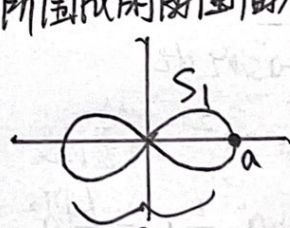
$$\begin{aligned} S_A &= \frac{1}{2} \int_0^{\frac{\pi}{4}} r^2 d\theta = \int_0^{\frac{\pi}{4}} (1 + \cos \theta)^2 d\theta \\ &= \frac{\pi}{8} + 2 \sin \theta \bigg|_0^{\frac{\pi}{4}} + \frac{1}{4} \int_0^{\frac{\pi}{4}} 1 + \cos 2\theta d\theta = \frac{1}{8} + \frac{\sqrt{2}}{2} + \frac{1}{16} \pi \end{aligned}$$

$$S = S_{ABC} - S_A$$

1000.10t.7

双纽线 $r^2 = a^2 \cos 2\theta$ ($a > 0$) 绕极轴旋转一周
所围成的曲面面积 S.

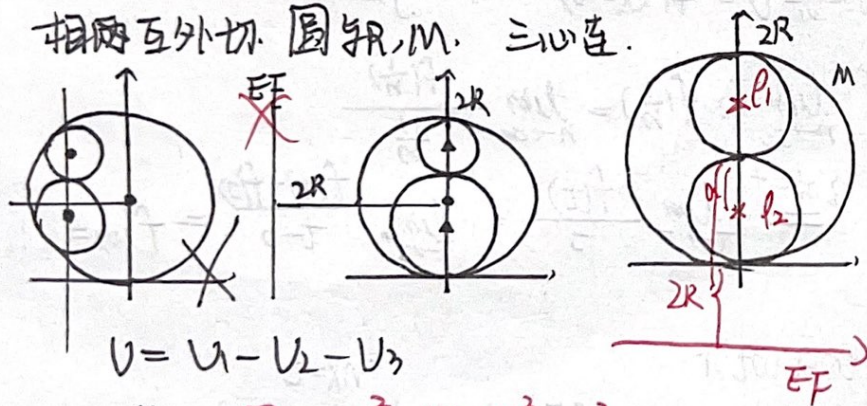
$$\begin{aligned} S &= 2S_1 = 2 \int_0^a y \sqrt{(x'_t)^2 + (y'_t)^2} dx \\ &= 2 \int_0^a a \sqrt{\cos 2\theta} \cdot \sqrt{(\cos 2\theta)^2 + \cos 2\theta} d\theta \\ &= 2 \int_0^a a^2 \sqrt{1 + \cos^2(2\theta)} \cdot \frac{1}{2} \cos \theta d\theta \\ x &= r \cos \theta = a \sqrt{\cos 2\theta} \cdot \cos \theta \\ y &= r \sin \theta = a \sqrt{\cos 2\theta} \cdot \sin \theta \end{aligned}$$



$$\begin{aligned}
 S &= 2S_1 = 2 \cdot \int_0^{\frac{\pi}{4}} 2\pi r \sin\theta \cdot \sqrt{r^2 + (r')^2} d\theta \\
 &= 4\pi \int_0^{\frac{\pi}{4}} a \sqrt{\cos 2\theta} \cdot \sin\theta \cdot \sqrt{a^2 \cos^2 2\theta + \left[a \frac{(-2)\sin 2\theta}{2\sqrt{\cos 2\theta}}\right]^2} d\theta \\
 &= 4\pi a^2 \int_0^{\frac{\pi}{4}} \sqrt{\cos 2\theta} \cdot \sin\theta \cdot \sqrt{\cos^2 2\theta + \left(\frac{\sin 2\theta}{\sqrt{\cos 2\theta}}\right)^2} d\theta \\
 &= 4\pi a^2 \int_0^{\frac{\pi}{4}} \sqrt{\cos 2\theta} \cdot \sin\theta \cdot \frac{1}{\sqrt{\cos 2\theta}} d\theta \\
 &= 4\pi a^2 \int_0^{\frac{\pi}{4}} \sin\theta d\theta = 2\pi a^2 (2 - \sqrt{2}).
 \end{aligned}$$

1000. 10t. 8

相外切. 圓半徑, M . 三心連.



$$V = V_1 - V_2 - V_3$$

通求.

$$\text{圓: } x^2 + (y - p)^2 = a^2$$

$$\begin{aligned}
 V_0 &= \pi \int_{-a}^a [(p + \sqrt{a^2 - x^2})^2 - (p - \sqrt{a^2 - x^2})^2] dx \\
 &= 8\pi p \int_0^a \sqrt{a^2 - x^2} dx = 2\pi^2 p a^2.
 \end{aligned}$$

$$V = V_1 - V_2 - V_3$$

$$= 2\pi (2R) R^2 - 2\pi (3R - r) r^2 - 2\pi^2 (2R - r)(R - r)^2$$

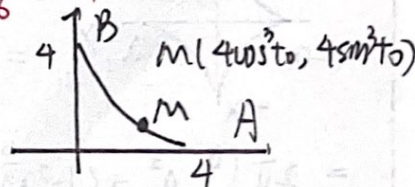
$$= 2\pi^2 (2R^3 - 7Rr^2 + 5R^2r) \quad (0 < r < R).$$

$$\frac{dV}{dr} = 0 \rightarrow r = \frac{7 - \sqrt{19}}{6} R$$

$$2R - r = \frac{\sqrt{19} - 1}{6} R$$

1000. 10t. 9

$$\begin{cases} x = 4\cos^3 t \\ y = 4\sin^3 t \end{cases} \quad t \in [0, \frac{\pi}{2}]$$



$$3 \int_{t_0}^{\frac{\pi}{2}} \sqrt{(x')^2 + (y')^2} dt$$

$$= 3 \int_{t_0}^{\frac{\pi}{2}} 4 \sqrt{(3\cos^2 t \cdot \sin t)^2 + (3\sin^2 t \cdot \cos t)^2} dt$$

$$= 12 \int_{t_0}^{\frac{\pi}{2}} \sqrt{\cos^4 t \sin^2 t + \sin^4 t \cos^2 t} dt = 12 \int_{t_0}^{\frac{\pi}{2}} \cos t \sin t dt$$

$$= 6 \int_{t_0}^{\frac{\pi}{2}} \sin 2t dt = -6 \frac{\cos 2t}{2} \Big|_{t_0}^{\frac{\pi}{2}} = -3(-1 - \cos 2t_0)$$

$$= 3 + 3\cos 2t_0$$

$$\int_{t_0}^{\frac{\pi}{2}} \sqrt{(x')^2 + (y')^2} dt = -3\cos 2t \Big|_{t_0}^{\frac{\pi}{2}} = -3(\cos 2t_0 - 3)$$

$$-3\cos 2t_0 = 3\cos 2t_0 - 3 \rightarrow -3\cos 2t_0 = 3 \rightarrow \cos 2t_0 = -1 \rightarrow 2t_0 = \pi \rightarrow t_0 = \frac{\pi}{2}$$

$$-3(\cos 2t_0 - 3) = \frac{1}{4} \cdot 6 = \frac{3}{2} \rightarrow \cos 2t_0 = \frac{1}{2} \rightarrow 2t_0 = \frac{\pi}{3} \rightarrow t_0 = \frac{\pi}{6}$$